SQL

```sql
-- operators
-- arithmetic operator
SELECT 10+10 as TOTAL ;

SELECT 20 -5 as TOTAL;

SELECT 20 *5 as TOTAL;
SELECT 20 /5 as TOTAL;

SELECT 17 % 5 as TOTAL;

-- comparison operator (relatioinal operators )
-- = , != or <> , > , <, >=, <=


SELECT 10=9 AS result;
SELECT 10 != 9 AS result;
SELECT 10 <> 9 AS result;
SELECT 10 > 10 AS result;
SELECT 10 < 10 AS result;
SELECT 10 >= 10 AS result;
SELECT 10 <= 10 AS result;

-- logical operators
SELECT 10=10 AND 9<>10 AS Result;
SELECT 10=10 OR 10<>10 AS Result;
SELECT  NOT(10=1) as Result;

-- asignment operator
-- =

-- special operator
-- Between, in , like. is null

-- types of sql commands
-- DDL (data definition language)
 -- used to create, modify, and delete database structures
 -- (databases, tables, schemas)

 CREATE DATABASE karthika;

 USE karthika;

 CREATE TABLE students(
    Sno INT,
```

```
    Name VARCHAR(200),
    Phone BIGINT
 );

 ALTER TABLE students
 ADD COLUMN dept VARCHAR(100);

 ALTER TABLE students
 CHANGE COLUMN dept Department VARCHAR(200);

 ALTER TABLE students
 RENAME TO student_details;

ALTER TABLE student_details
 DROP Department;

 DROP TABLE student_details;

 TRUNCATE TABLE student_details;



-- DML (data manipulation language)

INSERT INTO student_details VALUES(1, "Karthika", 9890348765);



-- DQL (data query language )
-- DCL (data control language)
-- TCL (Transaction control language)

INSERT INTO student_details VALUES(1, "Karthika", 9890348765);
INSERT INTO student_details VALUES(2, "Fiit", 965567755);
INSERT INTO student_details VALUES(3, "Academy", 7904908822) , (4, "Redmi",
8907893710);


UPDATE student_details
SET Name = "abcd";



UPDATE student_details
SET Name = "Karthika"
WHERE Sno=1;

DELETE FROM  student_details
WHERE Sno=4;
```

```sql
DELETE FROM  student_details

-- DQL (Data Query langauge)

SELECT * FROM student_details;

-- FROM -> SELECT

SELECT Phone , Name FROM student_details;
```

Theory Notes

1. SQL Operators

- **Arithmetic** → +, -, *, /, %.
- **Comparison (Relational)** → =, != or <>, >, <, >=, <=.
- **Logical** → AND, OR, NOT.
- **Special** → BETWEEN, IN, LIKE, IS NULL.
- **Assignment** → = (used in UPDATE).

2. SQL Commands

- **DDL (Data Definition Language)** → CREATE, ALTER, DROP, TRUNCATE.
- **DML (Data Manipulation Language)** → INSERT, UPDATE, DELETE.
- **DQL (Data Query Language)** → SELECT.
- **DCL (Data Control Language)** → GRANT, REVOKE.
- **TCL (Transaction Control Language)** → COMMIT, ROLLBACK, SAVEPOINT.

3. Table Operations

- CREATE TABLE → define structure.
- ALTER TABLE → add, modify, rename, or drop columns.
- DROP TABLE → delete table permanently.
- TRUNCATE TABLE → remove all rows but keep structure.
- INSERT INTO → add records.
- UPDATE → modify records.
- DELETE → remove records.
- SELECT → query records.

Coding Round Questions

1. Write a query to create a table employees with columns id, name, salary.
2. Insert 3 records into employees.
3. Update salary of employee with id=2.
4. Delete employee with id=3.
5. Select only name and salary from employees.
6. Write a query to find employees with salary greater than 50,000.
7. Demonstrate BETWEEN, IN, and LIKE operators with examples.

⚡ **Machine Coding Round Challenge**

Problem: Library Management Database

Requirements:

- Create a database LibraryDB.
- Create a table Books with columns: BookID, Title, Author, Price.
- Insert at least 5 books.
- Queries to implement:
  - Find all books priced between 200 and 500.
  - Update the price of a book with BookID=3.
  - Delete a book with BookID=5.
  - Select all books by a specific author.
  - Use LIKE to find books whose title starts with "Data".